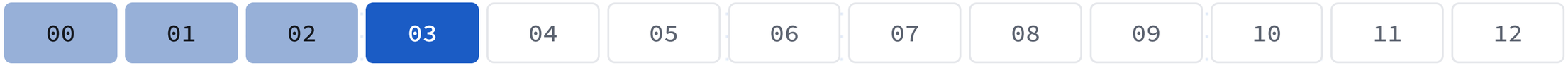
MODULE 03 · AMBITIOUS ENTREPRENEURSHIP

The Problem Space

Is this a real problem, or just an interesting one?

COURSE PATH



COURSE MATERIAL · SCAN NOW

All slides live on GitHub

The newest version, free, no account needed

github.com/BaNburger/
AmbitiousEntrepreneurship

1. **Open the link**
   github.com/BaNburger/AmbitiousEntrepreneurship
2. **Go to modules**
   then 03-problem-space
3. **Click the lecture PDF**
   M03_Lecture_Problem-Space.pdf, download top right

Public course repository of Ambitious Entrepreneurship. The QR code opens github.com/BaNburger/AmbitiousEntrepreneurship.

WHERE WE ARE · THE COURSE

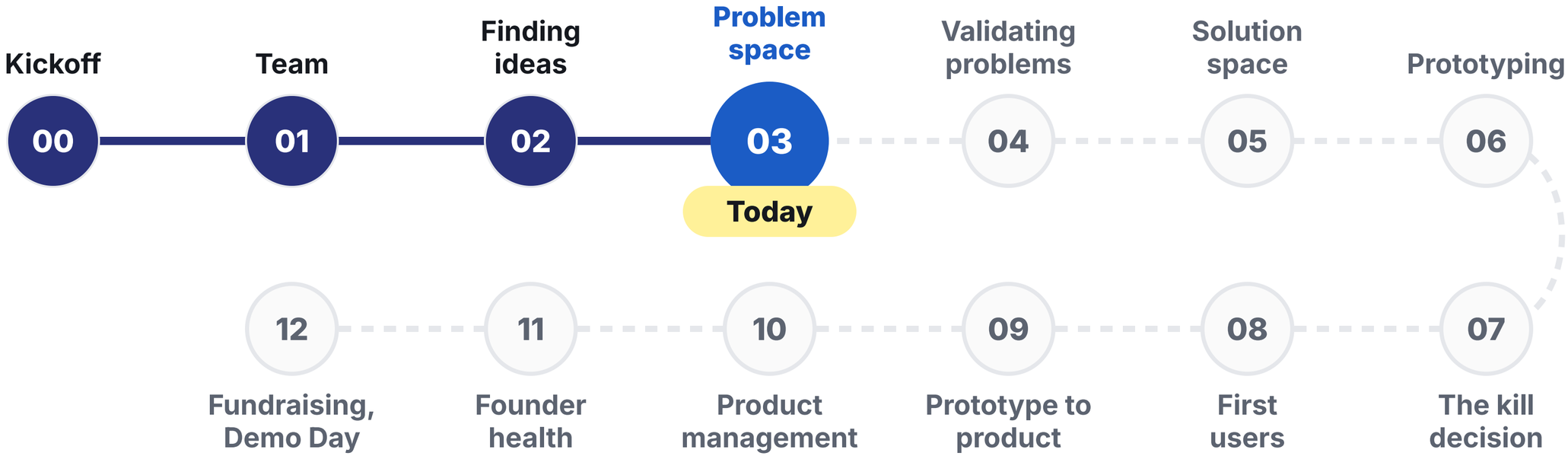
From one idea to a real problem

Thirteen sessions from first idea to first revenue. What teams have by now:

A team that committed
Team (M01)

One idea, picked on evidence
Finding ideas (M02)

Talks with named people
Finding ideas (M02)



CHECK-IN · SHOW OF HANDS

How many problem conversations did you hold since last time?

Hands up for your band. One click on the band per hand

0
not yet

1–4
started
under the floor of 5

5–14
floor reached

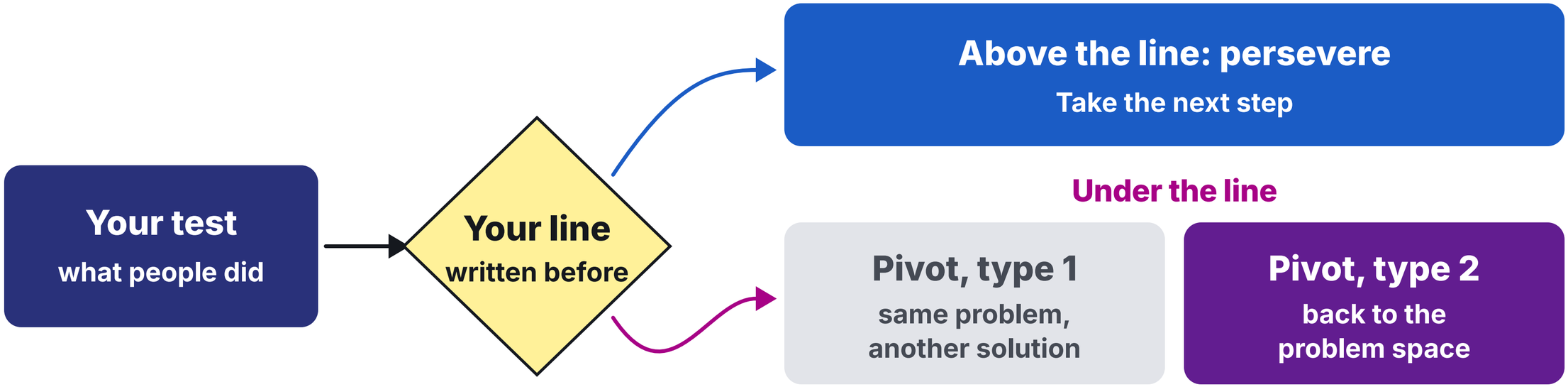
15+
full bar
15 per student

2:00

PIVOT OR PERSEVERE

Under your line? That is a decision, not a failure

Before there is a solution, a pivot goes back to the problem



Both are progress. The only miss: no line, so no decision

FRAMEWORK Pivot or persevere after Ries, E. (2011), The Lean Startup. Pivot types as in the course Glossary.

ACCOUNTABILITY

Report what you did, not what you planned

1

User interactions?

What did you learn?

2

New revenue?

Why did they pay?

3

Shipped?

Why does it matter?

🦄 **Report target vs. actual in small groups**

CHAPTER 1 OF 9

Previously on Ambitious Entrepreneurship

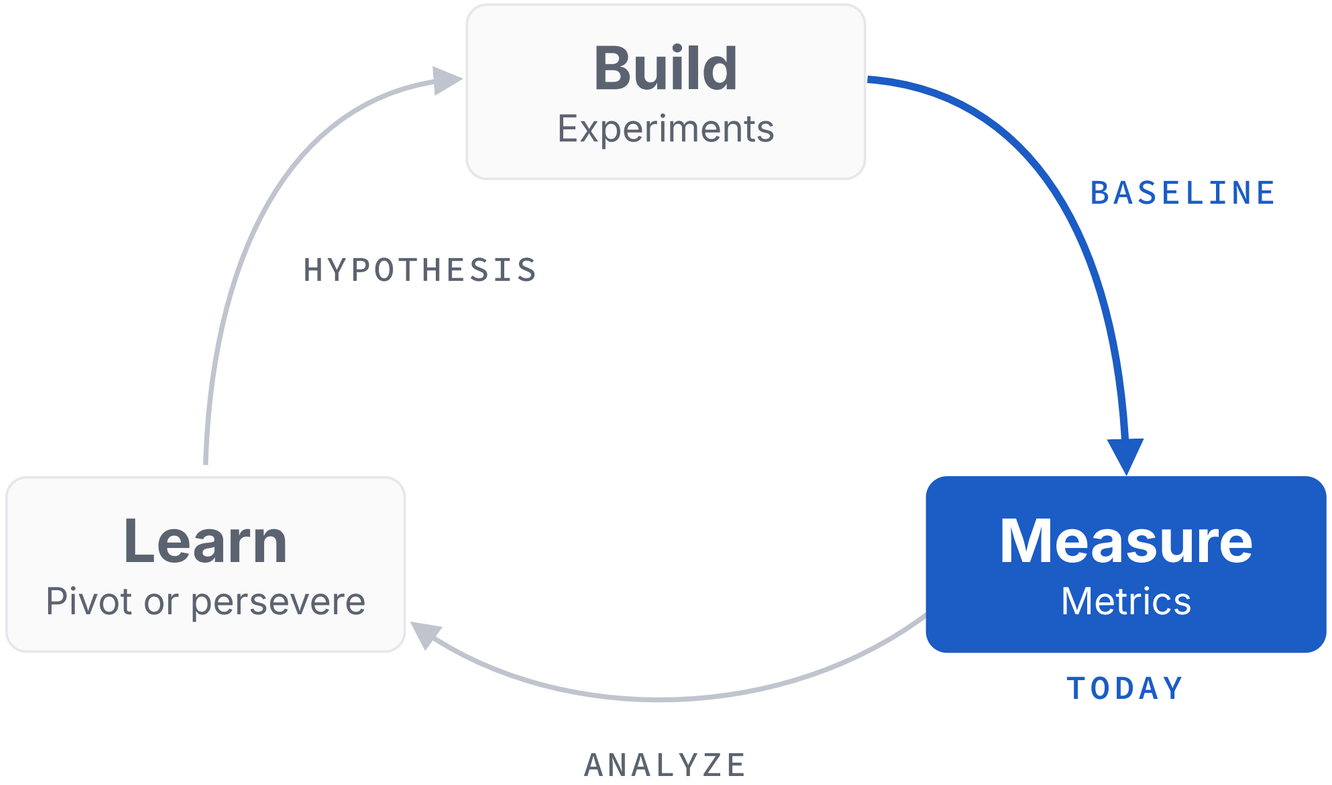
Two pictures from earlier sessions, then we test whether your idea's problem is real.

01 **Previously on Ambitious Entrepreneurship**
02 Your presentations
03 **Dig it: the problem behind your idea**
04 **Price it: the job and its workaround**
05 **Hear it: conversations about the last time**
06 **Pick it: the segment with its hair on fire**
07 Wrap-up and check-out
08 Workshop: find the pain worth fixing
09 Case discussion

PREVIOUSLY · KICKOFF (M00)

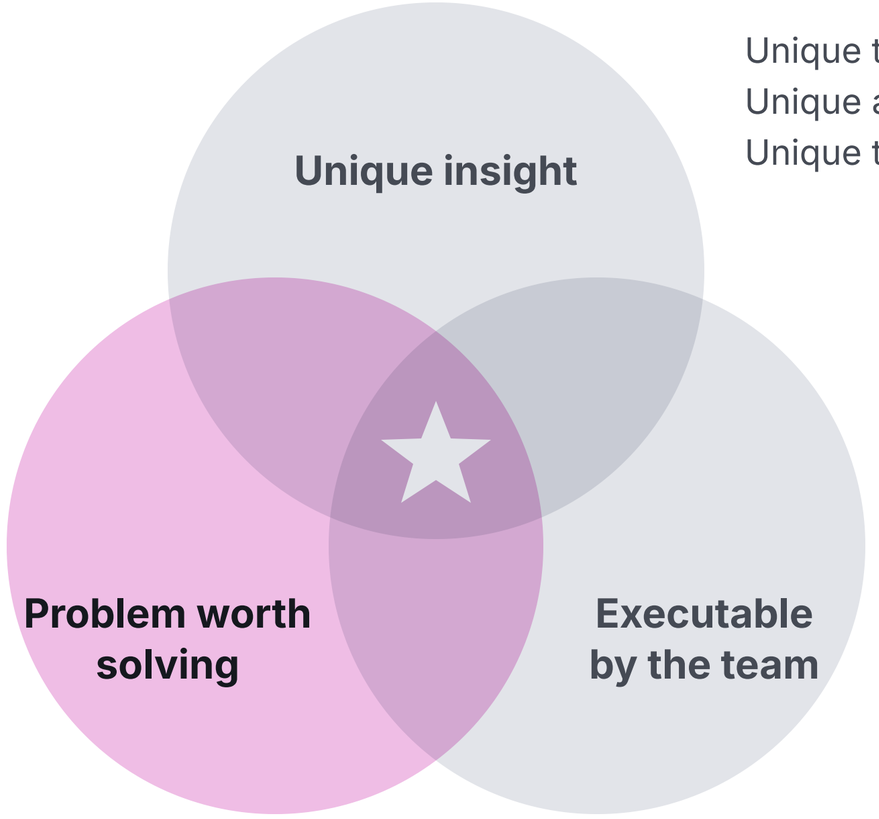
Your idea is a hypothesis

Today you measure the problem



PREVIOUSLY · FINDING IDEAS (M02)

Today we test worth solving



Unique tech
Unique access to customers
Unique take that others don’t have

You are passionate about it
Your audience is passionate about it
You can make a good living from it

You can scale the product
You can scale the backend
You can scale the sales

CHAPTER 2 OF 9

Your presentations

Each team shows in three minutes what it did, learned and needs from us.

01 ~~Previously on Ambitious Entrepreneurship~~

02 **Your presentations**

03 Dig it: the problem behind your idea

04 Price it: the job and its workaround

05 Hear it: conversations about the last time

06 Pick it: the segment with its hair on fire

07 Wrap-up and check-out

08 Workshop: find the pain worth fixing

09 Case discussion

YOUR PRESENTATIONS · MAKE IT CREDIBLE

Show us where you're at in 3 minutes or less!

3:00

Facts
Research on the problem, with sources

12 Numbers
Conversations per student, ideas dropped

Quotes
The three that decided it

Images
Your Ideas sheet, heard or assumed

Verdict
Kept or dropped at the kill line

Hypothesis
Your pitch in one to three sentences

Your target vs. actual | **What did you do?** | **What did you learn?** | **How can we help?**

INPUT 1 OF 4 · CHAPTER 3 OF 9

Dig it: the problem behind your idea

Describe the problem without your product, then dig until you reach the root cause.

01 ~~Previously on Ambitious Entrepreneurship~~

02 ~~Your presentations~~

03 **Dig it: the problem behind your idea**

04 **Price it: the job and its workaround**

05 **Hear it: conversations about the last time**

06 **Pick it: the segment with its hair on fire**

07 Wrap-up and check-out

08 Workshop: find the pain worth fixing

09 Case discussion

INPUT 1 OF 4 · DIG IT · THREE QUESTIONS

Before you build: what is the problem?

1

Why start with the problem, not the product?

2

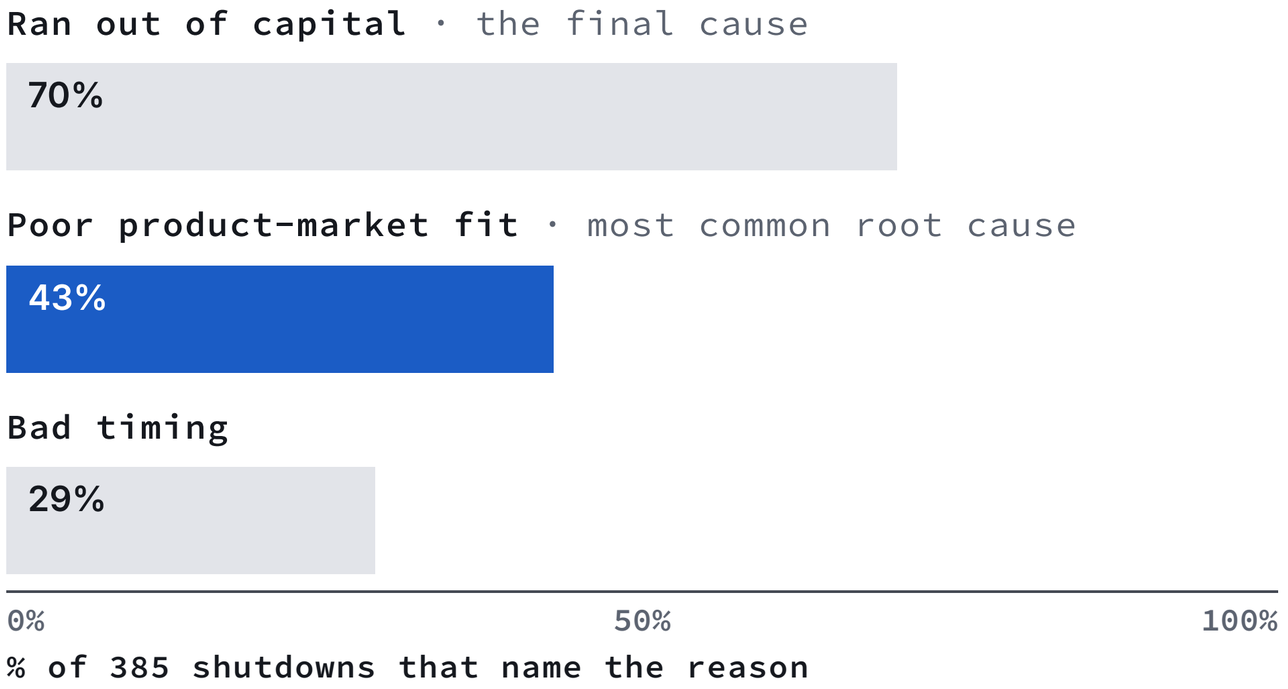
How do you describe a problem without your product?

3

How do you find its root cause?

WHY IT MATTERS

Money runs out last. Poor fit often broke first

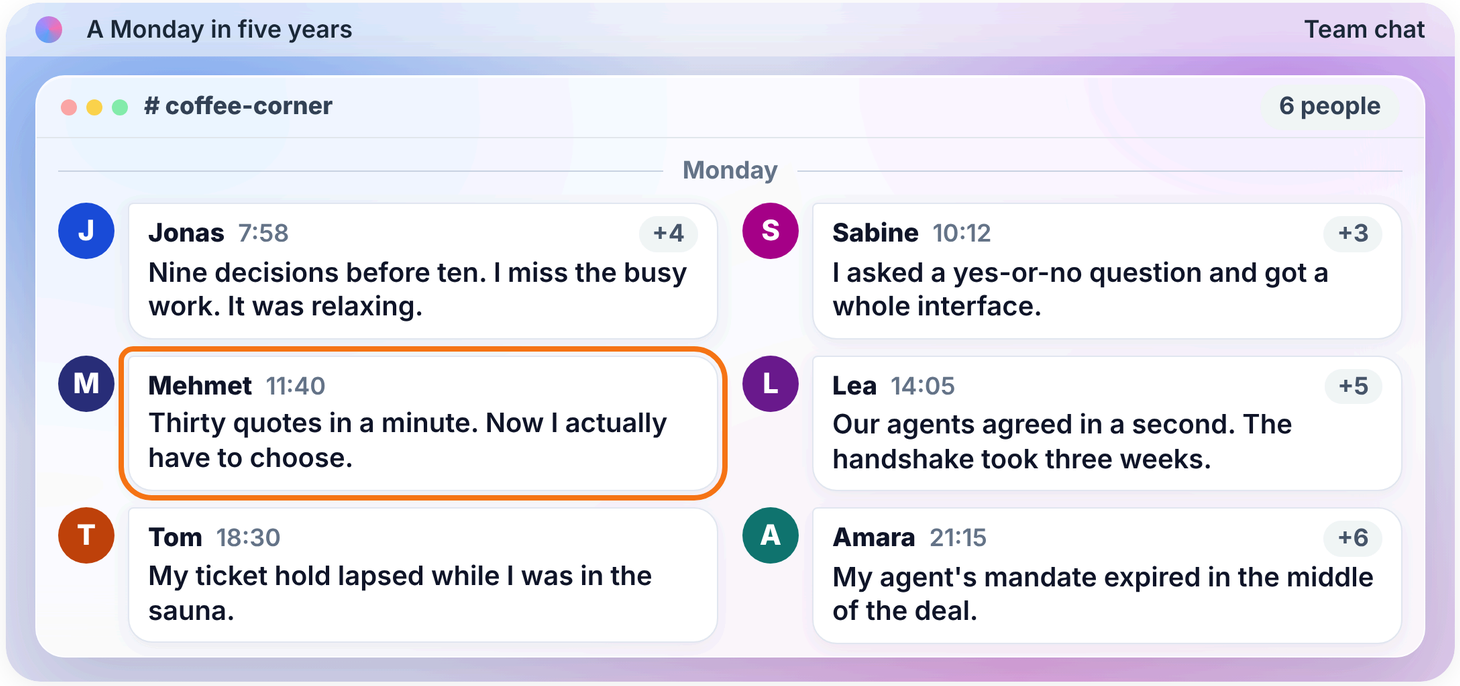


DATA CB Insights, The top 9 reasons startups fail, 5 March 2026. 385 VC-backed shutdowns since 2023 with known reasons, several reasons each. Bars to scale.

A DAY IN FIVE YEARS · SCENARIO

In five years, we still complain. About new things

Each complaint is a problem someone will have. Find the people who have it today



> "Thirty quotes in a minute. Now I actually have to choose."

↓

PROBLEM

Buyers struggle with choosing among dozens of quotes

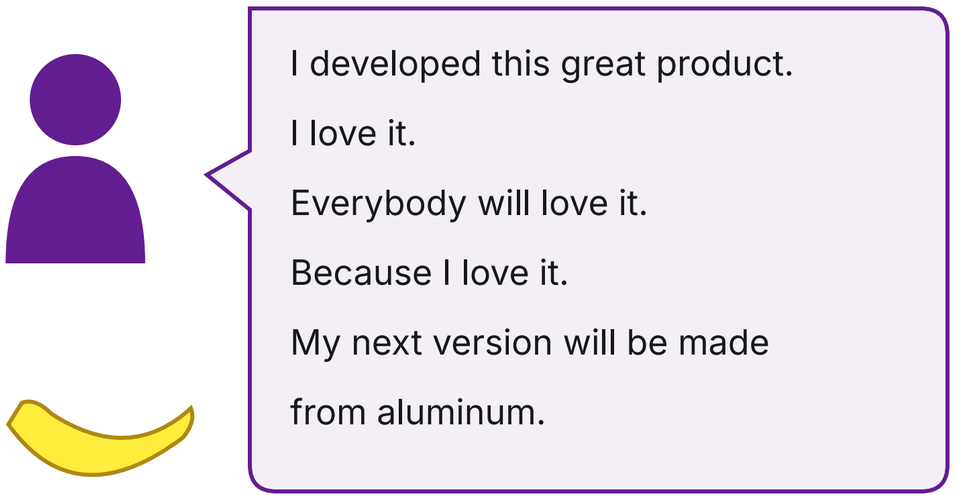
↓

Who has it today? Buyers who already compare dozens of bids. Call them

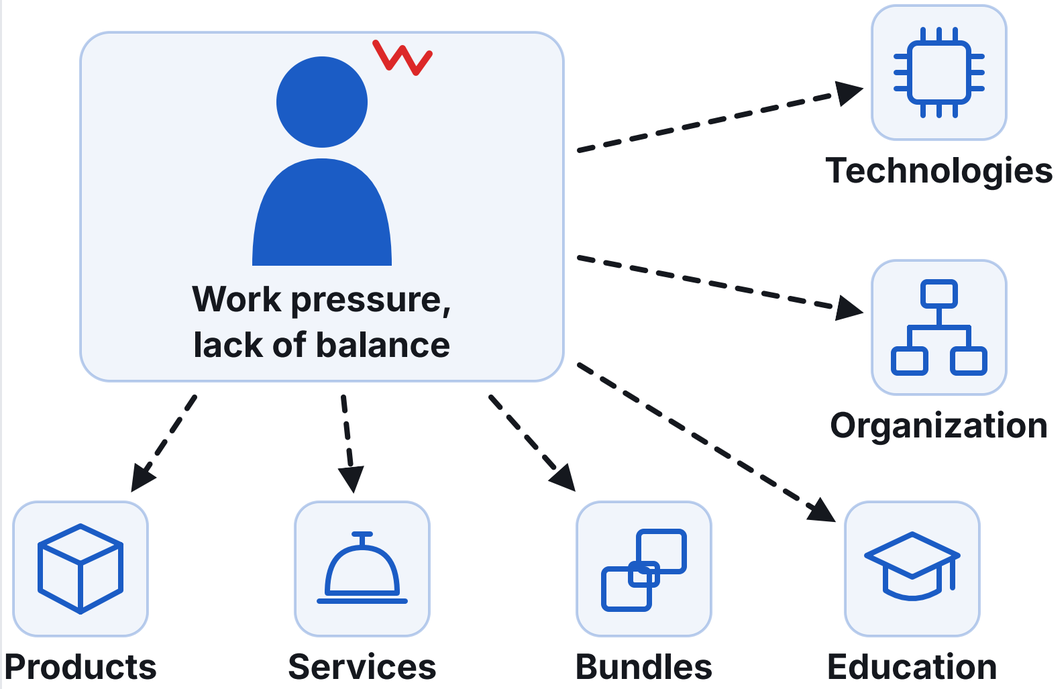
CORNERSTONE · PROBLEM ORIENTED

Love the problem, not your product

Solution oriented



Problem oriented



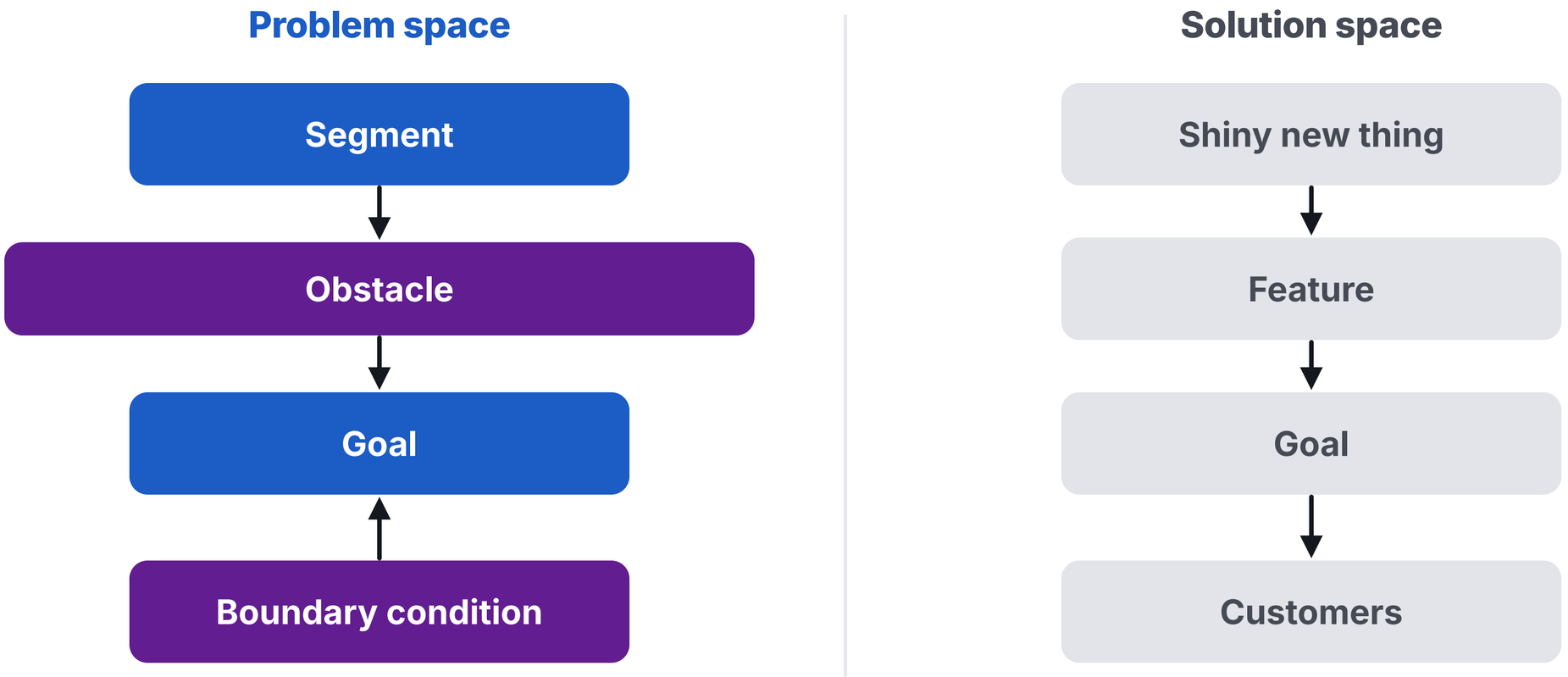
Start with a problem. Then look for solutions.

The other way round is a solution in search of a problem.

YC After Friedman, J. (2020). How to get startup ideas. YC Startup Library, recorded for Startup School 2020. "A solution in search of a problem" is his term.

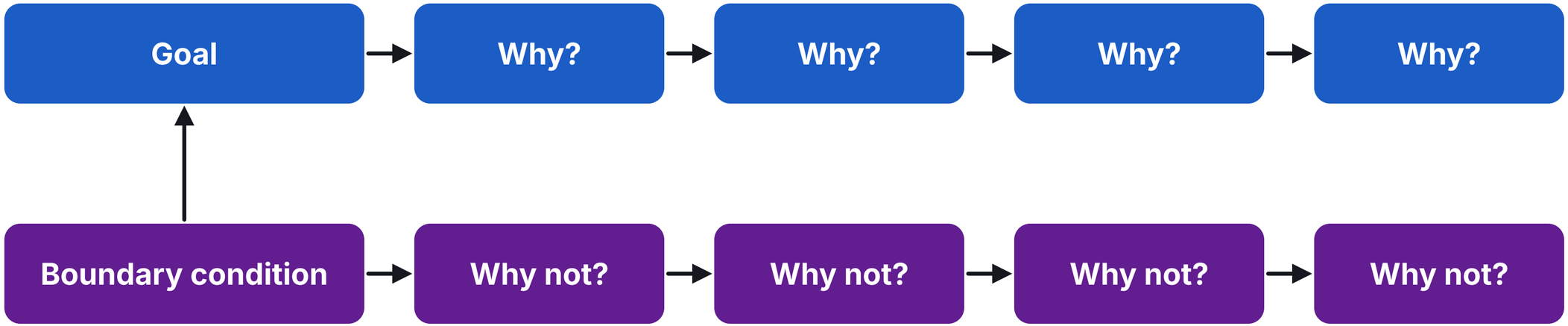
CORNERSTONE · PROBLEM SPACE VS. SOLUTION SPACE

Start left. Solutions come last



CORNERSTONE · FINDING ROOT CAUSES

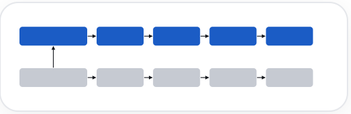
Ask why. Then ask why not



Goal The car starts every morning. Why?

Boundary No time for the garage. Why not?

FRAMEWORK 5 Whys after Ohno, T. (1988). *Toyota Production System*. Critique: Card, A. J. (2017). BMJ Quality & Safety. The car example is Illustrative.



EXAMPLE · OCEAN TRASH

The robot is a solution. Dig

Robot removes trash from the ocean

Most ocean trash comes from rivers

Mostly from dense cities

Awareness. Plastic is cheap.

EXAMPLE · THE CAR DOES NOT START

Fix the root, not the symptom

Symptom · Your car does not start

1st why
The alternator is not functioning → **Solution 1** → Feature 1, 2, 3 ...

2nd why
The alternator belt has broken → **Solution 2** → Feature 1, 2, 3 ...

3rd why
Belt worn out, never replaced → **Solution 3** → Feature 1, 2, 3 ...

Root cause
No maintenance on schedule → **Solution 4** → Feature 1, 2, 3 ...

Why not on schedule? No time for the garage

ILLUSTRATIVE The why-not row is added for this deck.



NOW YOU · ALONE · 4 MINUTES

Your problem space. No product words

Segment	Who has it worst?
Obstacle	What is in the way?
Goal	What do they want?
Boundary	What must any answer respect?
Why? ×3	Why does the obstacle exist?

4:00

OUTPUT

Four lines and a root cause

INPUT 1 OF 4 · DIG IT · THE ANSWERS

Dig it: the root, not the symptom

1

Why start with the problem, not the product?

Money runs out last. Poor fit, the most common root cause, often broke first

2

How do you describe a problem without your product?

Segment, obstacle, goal and boundary, with no product words

3

How do you find its root cause?

Ask why, then why not, until you reach the root, not the symptom

INPUT 2 OF 4 · CHAPTER 4 OF 9

Price it: the job and its workaround

Find the job people hire a product for and put a price on today's workaround.

01 ~~Previously on Ambitious Entrepreneurship~~

02 ~~Your presentations~~

03 ~~Dig it: the problem behind your idea~~

04 **Price it: the job and its workaround**

05 **Hear it: conversations about the last time**

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INPUT 2 OF 4 · PRICE IT · THREE QUESTIONS

What does the problem cost today?

1

What do people hire a product for?

2

What does the workaround cost them?

3

Why put a number on it?



FRAMEWORK · JOBS-TO-BE-DONE

They hire your product for a job

When [situation] → **I want to [motivation]** → **so I can [outcome]**

When I finish a day of appointments,

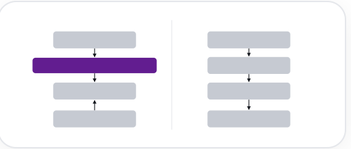
I want my notes done,

so I can go home without the screen.

FRAMEWORK Christensen, C. M., et al. (2016). Know your customers' "jobs to be done". *Harvard Business Review*. The physician's sentence is Illustrative, written for this deck.

EXAMPLE · PHYSICIANS AND THEIR NOTES

The workaround costs hours. Count them



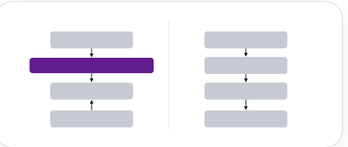
+ 1 to 2 hours
every night, self-reported by 21 of them

57
US physicians observed

RESEARCH Sinsky, C., et al. (2016). Allocation of physician time in ambulatory practice. *Annals of Internal Medicine*, 165(11). Bars to scale.

EXAMPLE · AN AI SCRIBE, ONE HEALTH SYSTEM

Your workaround's number is what a fix must beat



RESEARCH Duggan, M. J., et al. (2025). *JAMA Network Open*. One health system, 46 clinicians, 2024, before and after, not randomized. Each bar is that row's time before, to scale.



NOW YOU · ALONE · 3 MINUTES

Write the job. Price the workaround

When [situation]

I want to [motivation]

so I can [outcome]

It costs how many euros or hours?

3:00

OUTPUT

One sentence, one number. Heard or assumed?

INPUT 2 OF 4 · PRICE IT · THE ANSWERS

Price it: the number a fix must beat

1

What do people hire a product for?

A job: when [situation], I want to [motivation], so I can [outcome]

2

What does the workaround cost them?

Count it in hours or euros, like the physicians' screen and desk time

3

Why put a number on it?

The workaround's number is what a fix must beat

Hear it: conversations about the last time

Ask about the last time it happened, never pitch, and keep their exact words.

01 ~~Previously on Ambitious Entrepreneurship~~

02 ~~Your presentations~~

03 ~~Dig it: the problem behind your idea~~

04 ~~Price it: the job and its workaround~~

05 **Hear it: conversations about the last time**

06 **Pick it: the segment with its hair on fire**

07 Wrap-up and check-out

08 Workshop: find the pain worth fixing

09 Case discussion

INPUT 3 OF 4 · HEAR IT · THREE QUESTIONS

How do you hear the truth?

1

Why not ask what people would do?

2

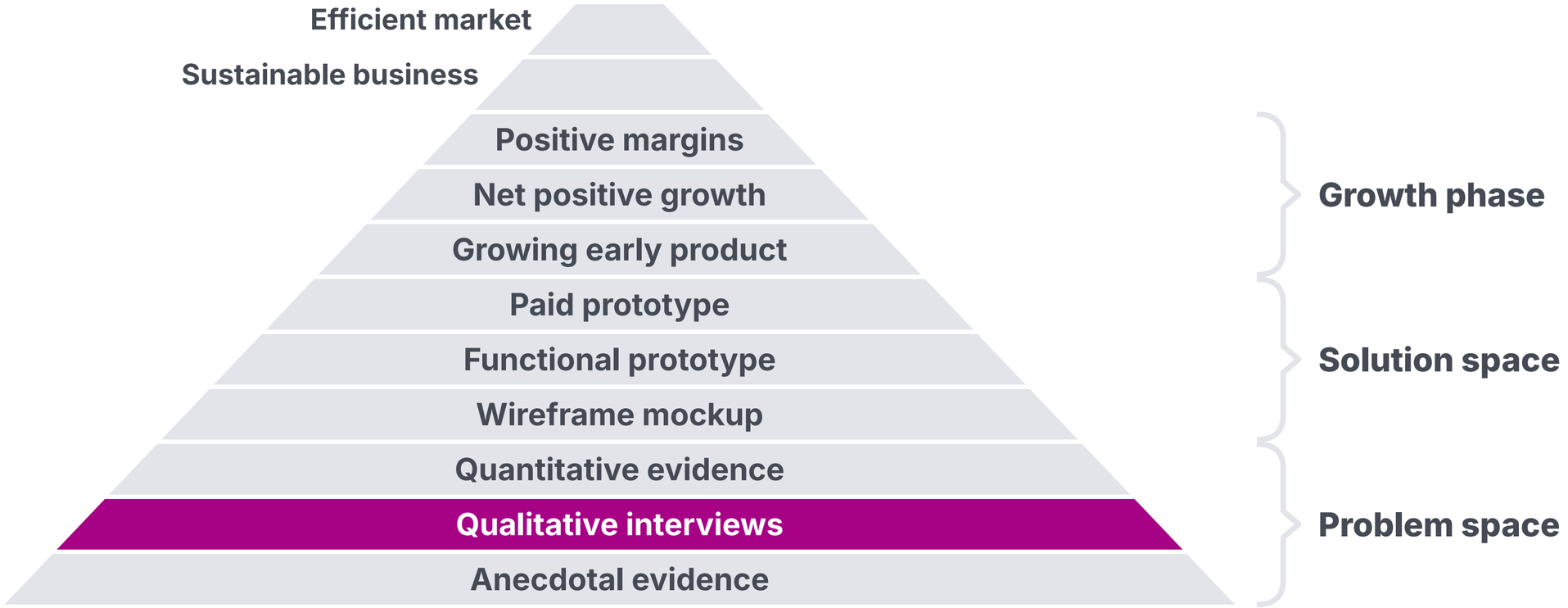
What do you ask instead?

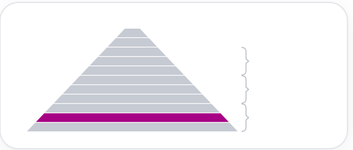
3

How many conversations are enough?

CORNERSTONE · HIERARCHY OF EVIDENCE

From today you stand on rung two





WHY IT MATTERS

What people plan is not what they do

Intentions moved this much · medium to large

d = 0.66

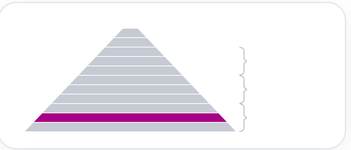
Behaviour moved about half as much · small to medium

d = 0.36

0 0.33 0.66

Average change, effect size d, 47 experiments

RESEARCH Webb & Sheeran (2006), 47 experiments that changed intentions. Effect sizes (d) to scale.



PROBLEM CONVERSATIONS · TIP 1 OF 4

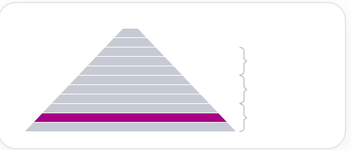
Ask about the last time, not in general

FROM

"What do you feel when shopping online?"

TO

"What did you feel the last time?"



PROBLEM CONVERSATIONS · TIP 2 OF 4

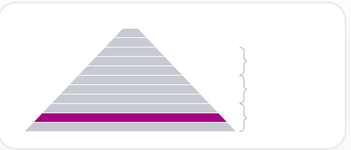
Don't lead. Discover

FROM

"Wouldn't an app make that easier?"

TO

"What do you do when it happens?"



PROBLEM CONVERSATIONS · TIP 3 OF 4

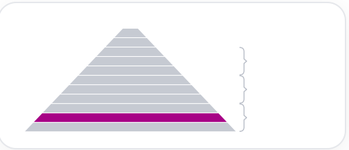
Never pitch. Listen

FROM

"Will you use our product?"

TO

"How do you solve it today?"



PROBLEM CONVERSATIONS · TIP 4 OF 4

Record it. Keep their exact words

YOUR SUMMARY

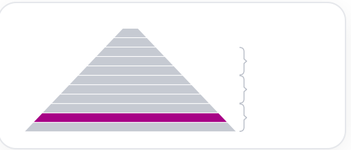
"annoying, wants automation"

THEIR WORDS

"Every Friday I copy 200 orders into Excel by hand."

Ask first: "May we record this for our team and the course staff, who may spot-check it?"



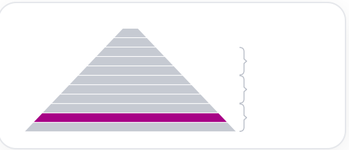


THE FIVE GOLDEN QUESTIONS

Ask these five, last time first

1. **Tell me about the last time…**
2. **What was the hardest part?**
3. **Why was that hard? Up to 5×**
4. **How do you solve it currently?**
5. **What don't you love about that solution?**

YC After Migicovsky (2019), based on Fitzpatrick (2013), *The Mom Test*. Order as in the course workshop.



NOW YOU · IN PAIRS · 3 MINUTES

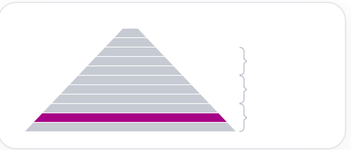
Fix your worst question

1. Take the worst question you asked or planned
2. Rewrite it as one of the five
3. Read it to your partner. Leading?

3:00

OUTPUT

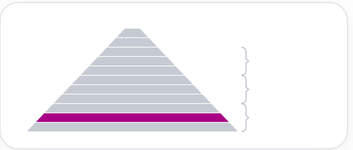
One question you can ask tomorrow



PROCESS · ONE CONVERSATION

End every conversation with a next step

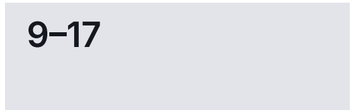
Gather intel	Set the stage	Interview	Follow through
Who are they? Their company?	Thank them. Time and goal.	Dig deep. A co-founder notes.	Who else? May we come back?



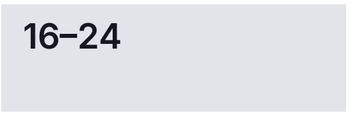
HOW MANY CONVERSATIONS

About 20 per segment

9 to 17 · narrow question, similar people



16 to 24 · to understand why



20 of 30 · found over 90% of the needs



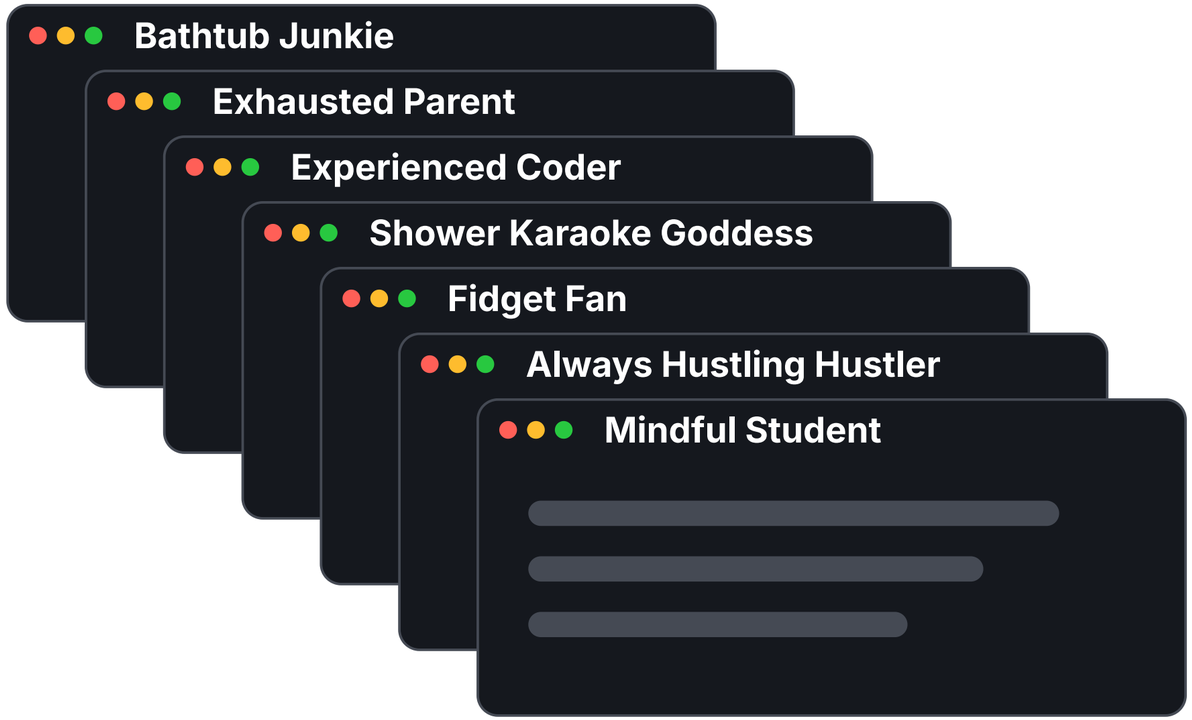
15 each

a team of three holds 45 in one week

RESEARCH Hennink & Kaiser (2022). Hennink et al. (2017). Griffin & Hauser (1993). Interviews to scale, 0 to 30.

AI REHEARSAL

Personas are sparring partners, not customers



Note

Use AI to understand problems, not to find solutions

AI REHEARSAL · WHY THEY NEVER COUNT

Personas match the average, miss the rest

Persona answers against a real survey



Differed from real people

48%

Flipped the sign · about 1 in 6

15%

0% 50% 100%

% of all persona estimates

Willingness to pay for fluoride toothpaste

Humans

1×

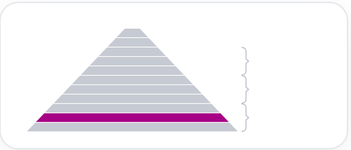
GPT

about 3×

0 1× 2× 3×

Premium, humans = 1×

RESEARCH Bisbee et al. (2024), *Political Analysis*. Brand, Israeli & Ngwe (2026), HBS Working Paper 23-062. Each group drawn to scale. The 15% are the 32% of the 48% that flipped.



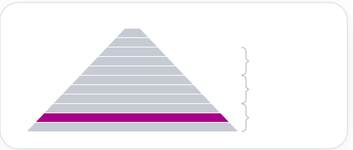
HOW-TO · REHEARSE WITH AI · STEP 1 OF 4

Open a fresh chat



1. **Open a fresh chat**
2. Paste the critical-customer prompt
3. Let it draft Mom Test questions
4. Rehearse until nobody says yes

Temporary chat: no memory.



HOW-TO · REHEARSE WITH AI · STEP 2 OF 4

Paste the critical-customer prompt

You

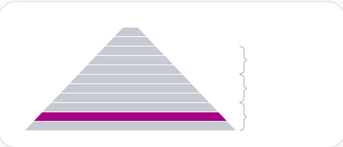
I am building a service in the field of {insert field here}. You are a potential critical customer of the service I build. You have extensive experience with competing services in the field. I will ask you questions about your situation, attitude and experience with the problem field to get to a better understanding of this field. You will answer critically, honest and direct to the questions and challenge my assumptions were necessary.

ChatGPT

Absolutely, I'm ready to assist you in this role-playing scenario as a critical customer with extensive experience in your chosen field. Please specify the field or industry of your service, and feel free to ask any questions related to customer attitudes, experiences, or challenges within that field. I'll provide critical, honest, and direct feedback, and challenge your assumptions where necessary.

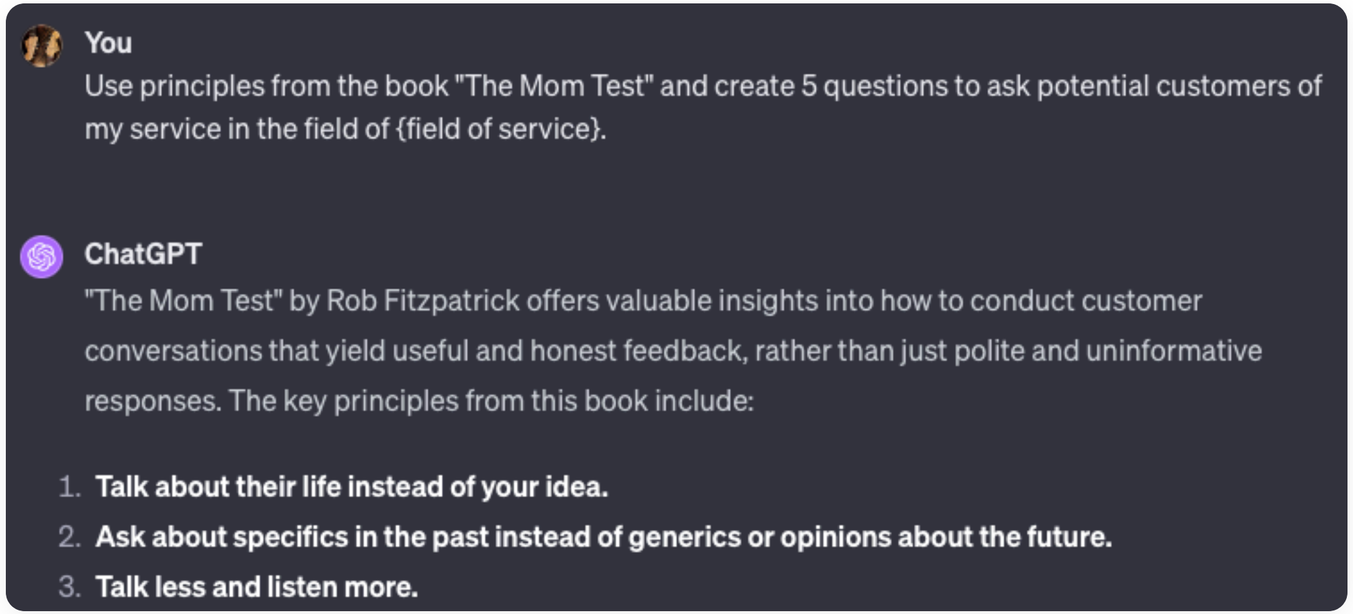
1. Open a fresh chat
2. **Paste the critical-customer prompt**
3. Let it draft Mom Test questions
4. Rehearse until nobody says yes

Role, task, iterate. Be specific.

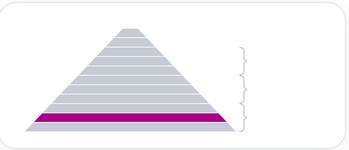


HOW-TO · REHEARSE WITH AI · STEP 3 OF 4

Let it draft Mom Test questions



1. Open a fresh chat
2. Paste the critical-customer prompt
3. **Let it draft Mom Test questions**
4. Rehearse until nobody says yes



HOW-TO · REHEARSE WITH AI · STEP 4 OF 4 · IN PAIRS

Rehearse until nobody says yes

1. Take the three it drafted, or three of yours
2. Rewrite each about one past event
3. Ask a persona. A yes means leading.

4:00

OUTPUT

Three open questions

INPUT 3 OF 4 · HEAR IT · THE ANSWERS

Hear it: the last time, in their words

1

Why not ask what people would do?

What people plan is not what they do. Behaviour moved half as much

2

What do you ask instead?

About the last time it happened: the five golden questions, never a pitch

3

How many conversations are enough?

About 20 per segment, each ending with a next step

Pick it: the segment with its hair on fire

Pick the segment that hurts most, often, and can pay, based on what you heard.

01 ~~Previously on Ambitious Entrepreneurship~~

02 ~~Your presentations~~

03 ~~Dig it: the problem behind your idea~~

04 ~~Price it: the job and its workaround~~

05 ~~Hear it: conversations about the last time~~

06 **Pick it: the segment with its hair on fire**

07 Wrap-up and check-out

08 Workshop: find the pain worth fixing

09 Case discussion

INPUT 4 OF 4 · PICK IT · THREE QUESTIONS

Whose hair is on fire?

1

What makes a segment worth picking?

2

Who decides, and who pays?

3

What do you base the pick on?

Their hair is on fire. Hand them a brick and they still grab it.

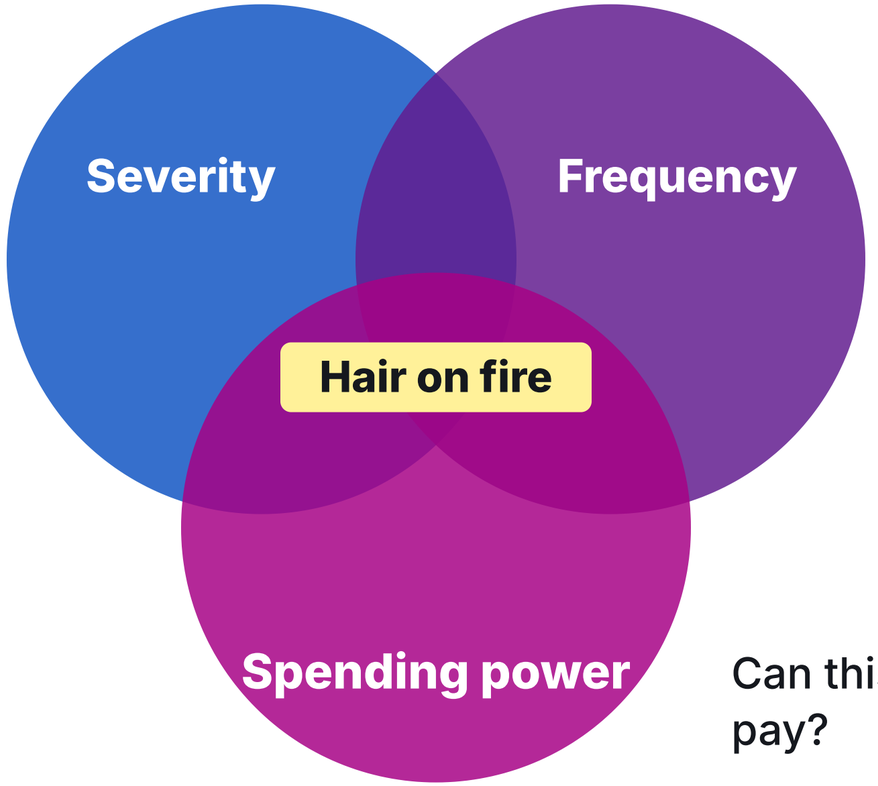
That segment forgives your first version. Start there.

YC After Seibel, M. (2016, June 27). The real product market fit. Y Combinator blog. The brick image is his, crediting Sequoia.

FRAMEWORK · THREE CIRCLES

It hurts, it happens often, they can pay

Money earned or saved?



How often?

Can this person pay?

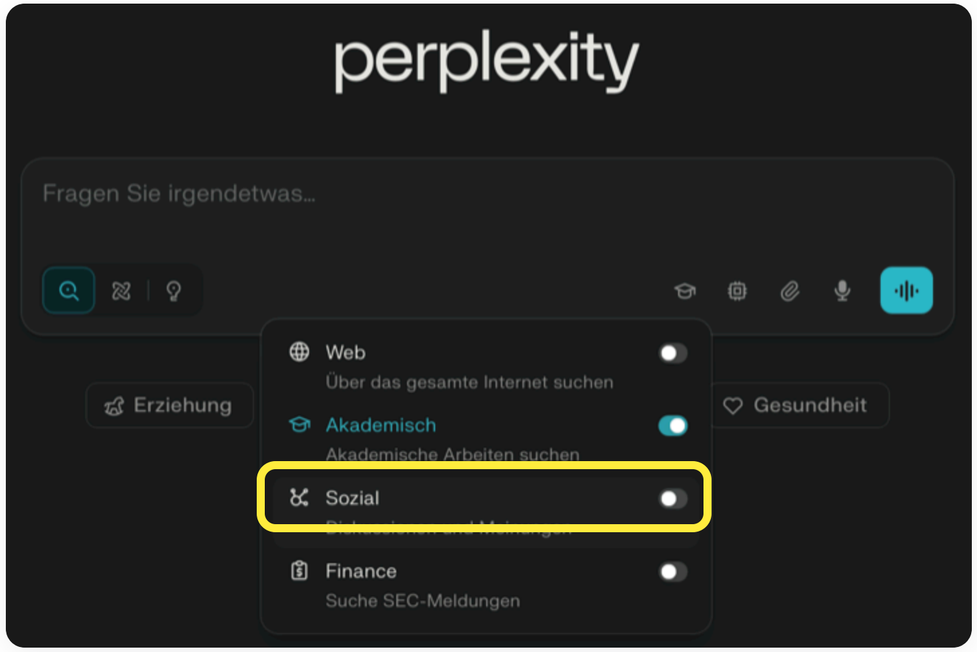
EXAMPLE · IMPLANTS IN US HOSPITALS

The chooser is not the payer



HOW-TO · AI DESK RESEARCH

Let AI find where they talk



1. **Pick sources, Social for forums**
2. **Ask where they talk and pay**
3. **Open every source**
4. **Message a named person**



YOUR PAIN MAP

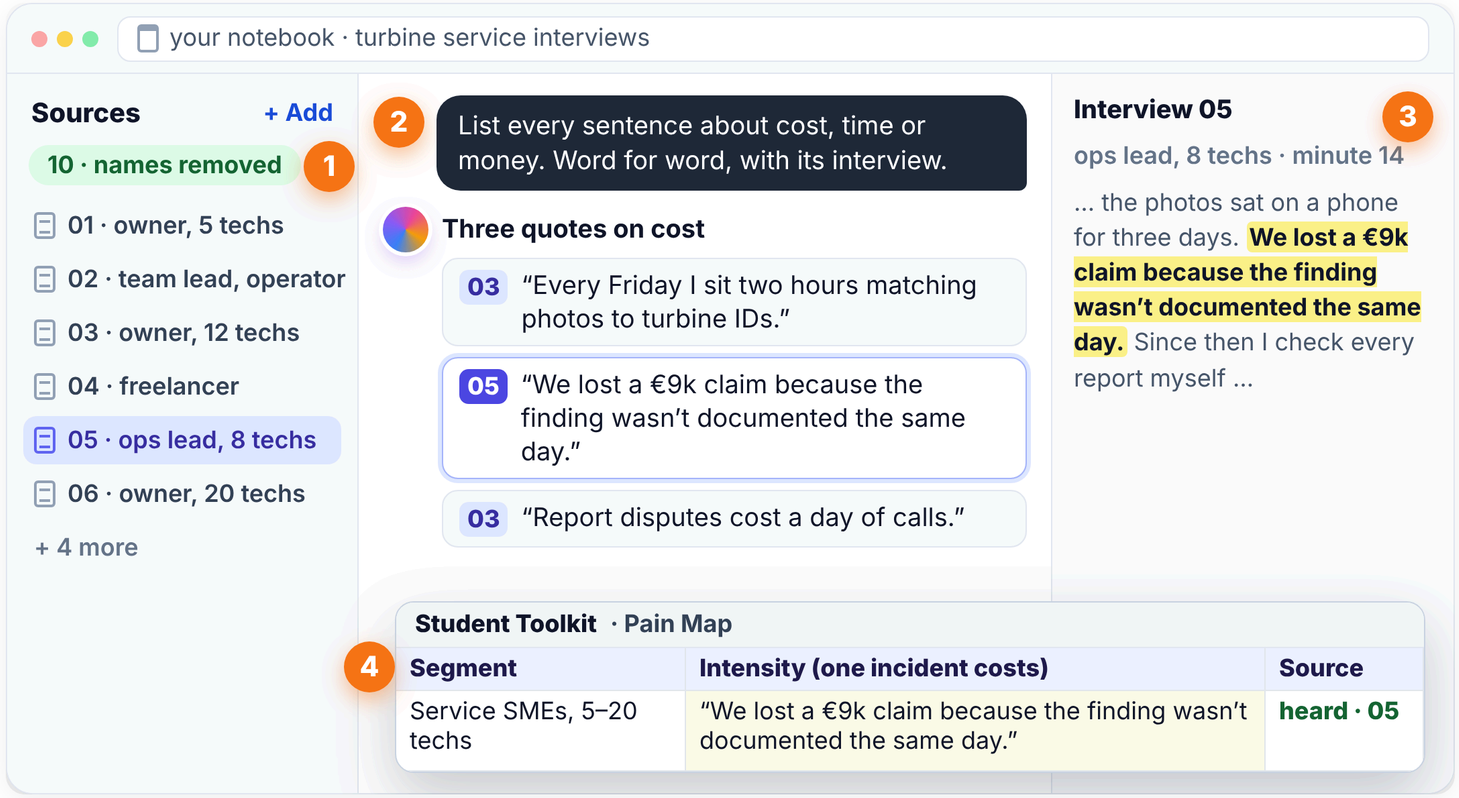
Commit on what you heard

	Frequency How often?	Intensity Cost of one time?	Current spend Paid today?	Budget holder Who signs?
Segment A	● **monthly** heard	○ **30 min** assumed	● **nothing** heard	○ **unclear** assumed
Segment B	● **daily** heard	● **2 hours** heard	● **€200 a month** heard	● **team lead** heard
Segment C	○ **weekly** assumed	● **1 hour** heard	○ **Excel only** assumed	○ **owner** assumed

ILLUSTRATIVE Pain Map columns as in the course workshop.

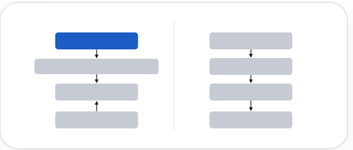
HOW-TO · FROM TRANSCRIPTS TO YOUR PAIN MAP

Let a notebook pull out the quotes



1. **Remove names, add transcripts**
2. **Ask for every quote on cost**
3. **Check each quote in its source**
4. **Paste it into your Pain Map**

ILLUSTRATIVE Screen prototyped for this course, quotes made up. NotebookLM is now Gemini Notebook (Google blog, 16 July 2026).



NOW YOU · AS A TEAM · 3 MINUTES

Name your segment and who signs

Who has it worst?

Who signs can say yes to paying?

Heard or assumed?

3:00

OUTPUT

A candidate, not a commitment

INPUT 4 OF 4 · PICK IT · THE ANSWERS

Pick it: it hurts, it's often, they pay

1

What makes a segment worth picking?

It hurts, it happens often, and they can pay

2

Who decides, and who pays?

Not always the same person: the surgeon chooses, the hospital pays

3

What do you base the pick on?

On what you heard, logged in your Pain Map, not what you assumed

CHAPTER 7 OF 9

Wrap-up and check-out

Look back at the cornerstones, reflect on the session and take your bar home.

01 ~~Previously on Ambitious Entrepreneurship~~

02 ~~Your presentations~~

03 ~~Dig it: the problem behind your idea~~

04 ~~Price it: the job and its workaround~~

05 ~~Hear it: conversations about the last time~~

06 ~~Pick it: the segment with its hair on fire~~

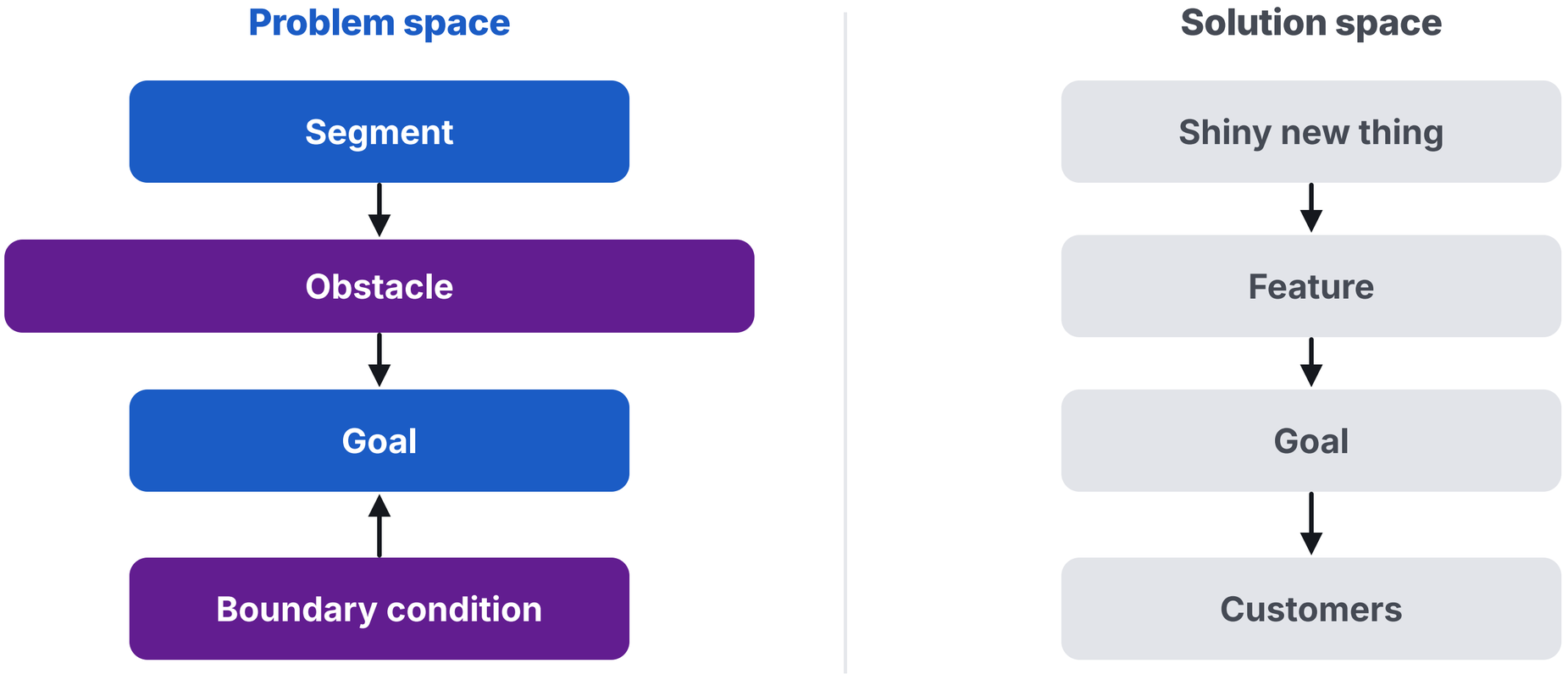
07 **Wrap-up and check-out**

08 Workshop: find the pain worth fixing

09 Case discussion

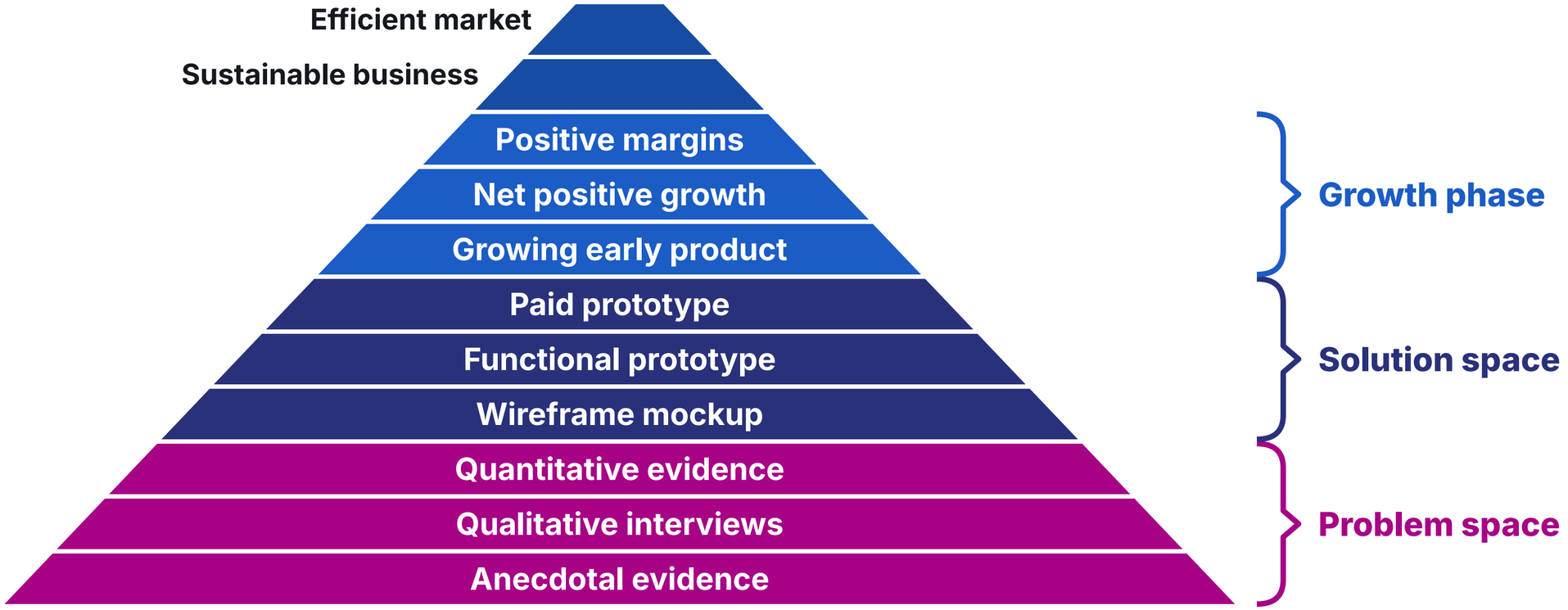
RECAP · CORNERSTONE

Stay left until you hear the cost



RECAP · CORNERSTONE

Evidence gets stronger near money



REFLECTION 1 OF 3 · THE CONTENT

What could you apply?



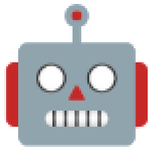
What could you apply?



What gave you a hard time?

REFLECTION 2 OF 3 · THE PROCESS

Man vs. machine



Where did AI actually help you?



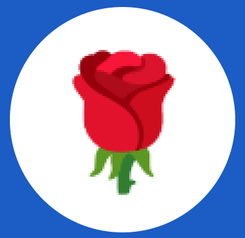
Where was your genius needed?



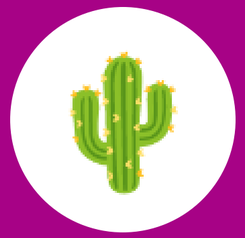
What are your best hacks to share?

REFLECTION 3 OF 3 · THE RESULTS

Rose, thorn, bud



What was the breakthrough today?



What was the hardest part today?



What is an idea/feature you didn't have time to finish, but want to explore later?

HOMEWORK UNTIL NEXT TIME

Hold 15 problem conversations per student in your hair-on-fire segment and come back with 5 named people per student who agreed to a next step.

15 each

The learning floor is 5 conversations and 2 next steps per student.

Log your kill line in the Toolkit's Kill Criteria Log before your first call.

HOMEWORK · HOW TO DO IT

Seven moves for your 15 conversations

1 **Go where they complain.**

2 **Open on their day.**

Watch out. Name the problem first, and it isn't unprompted.

3 **Never pitch.**

4 **Follow the money.**

5 **End with an ask, like a second call.**

6 **Record with the consent sentence in Legal Basics.**

7 **Get a blind second opinion.**

Watch out. AI personas are rehearsal and never count.

HOMEWORK · WHAT GOOD LOOKS LIKE

Come back with your first prospects

	Weak	Good
SEGMENT	"Companies"	**Solo HR managers, 50 to 200 staff**
UNPROMPTED	Only when named	**4 of 10**
WORKAROUND	"Annoying"	**5 hours a week, €300 a month**
NEXT STEPS	20 LinkedIn likes	**6 second calls booked**

ILLUSTRATIVE Problem space (M03) homework, what you come back with. A made-up example.

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See you next time!

MODULE 03 · WORKSHOP · 90 MINUTES

Workshop: Find the pain worth fixing

Build your Pain Map, hold a real conversation and leave with a week already moving.

Is this a real problem, or just an interesting one?

01 Previously on Ambitious Entrepreneurship

02 Your presentations

03 **Dig it: the problem behind your idea**

04 **Price it: the job and its workaround**

05 **Hear it: conversations about the last time**

06 **Pick it: the segment with its hair on fire**

07 Wrap-up and check-out

08 **Workshop: find the pain worth fixing**

09 Case discussion

WORKSHOP · STEP 1 OF 3

Define the pain, not the solution

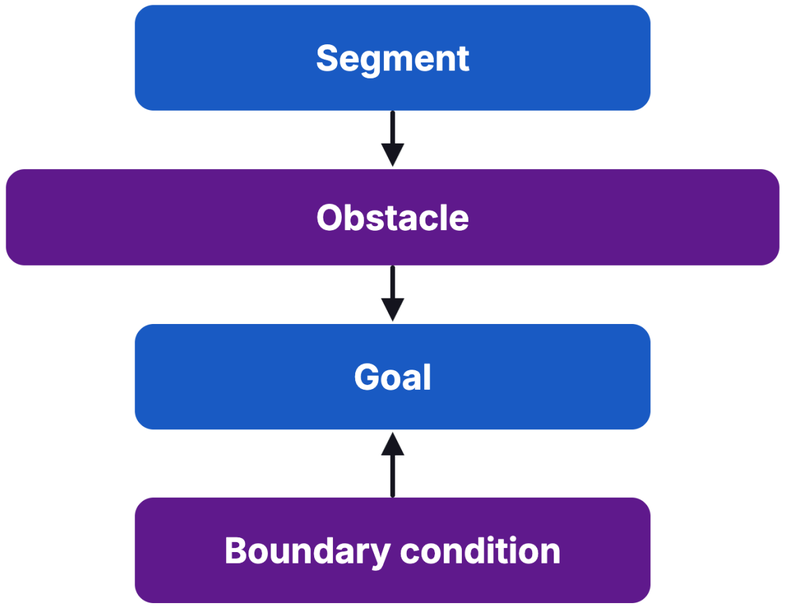
1 STEP 1 · 30 MIN

DEFINE THE PAIN, NOT THE SOLUTION

Agree on who hurts most, when, and at what cost

- → The segment: who suffers most?
- → The trigger: when does it hurt?
- → The fail state: what now, at what cost?
- → Job sentence, obstacle, boundary conditions, Pain Map, each cell heard or assumed
- → Kill line: result, date, consequence, logged before any call

Output Job sentence, Pain Map, hair-on-fire candidate, kill line **Minutes** 30



30:00

WORKSHOP · STEP 2 OF 3

The 5 golden questions, live

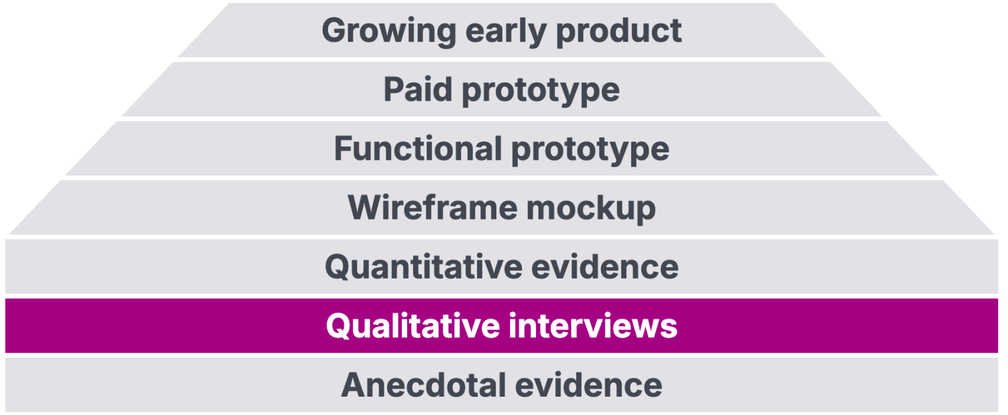
2 STEP 2 · 35 MIN

THE 5 GOLDEN QUESTIONS, LIVE

Is the problem expensive enough to fix? Do NOT pitch.

- → 10 min: open on their day, let them raise it first
- → Last time? Hardest part? Why ×5? Solved how? What don't you love?
- → End with a concrete ask
- → 25 min: one real conversation per team, logged

Output Script with ask, one conversation logged **Minutes** 35



35:00

WORKSHOP · STEP 3 OF 3

Move fast, then commit

3 STEP 3 · 25 MIN

MOVE FAST, THEN COMMIT

Leave with a week that is already moving

- → Deep research and social listening find where they talk
- → Named people, not personas
- → First messages sent before you leave
- → Each student states floor or full bar
- → Midweek day and pace

Output Target list, outreach sent, targets, midweek pace **Minutes** 25



25:00

ARRIVED EMPTY-HANDED?

No interviews yet? Hold two real conversations in Step 2.

2 real conversations

HOMEWORK UNTIL NEXT TIME

Hold 15 problem conversations per student in your hair-on-fire segment and come back with 5 named people per student who agreed to a next step.

15 each

The learning floor is 5 conversations and 2 next steps per student.

Log your kill line in the Toolkit's Kill Criteria Log before your first call.

See you next time!

MODULE 03 · CASE DISCUSSION · ABOUT 90 MINUTES

Case discussion: Konux

Stand in Konux's shoes in October 2016 and choose railway switches or several industries.

Is this a real problem, or just an interesting one?

01 Previously on Ambitious Entrepreneurship

02 Your presentations

03 **Dig it: the problem behind your idea**

04 **Price it: the job and its workaround**

05 **Hear it: conversations about the last time**

06 **Pick it: the segment with its hair on fire**

07 Wrap-up and check-out

08 Workshop: find the pain worth fixing

09 **Case discussion**

CASE · KONUX, MUNICH, 31 OCTOBER 2016 · VOTE 1 OF 2

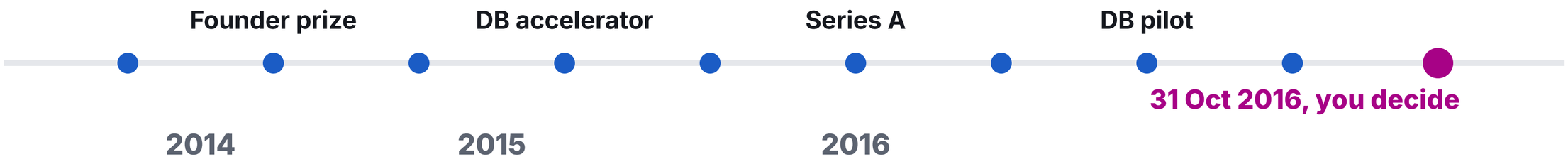
Railway switches or several industries?

OPTION A

Build Konux around railway switches

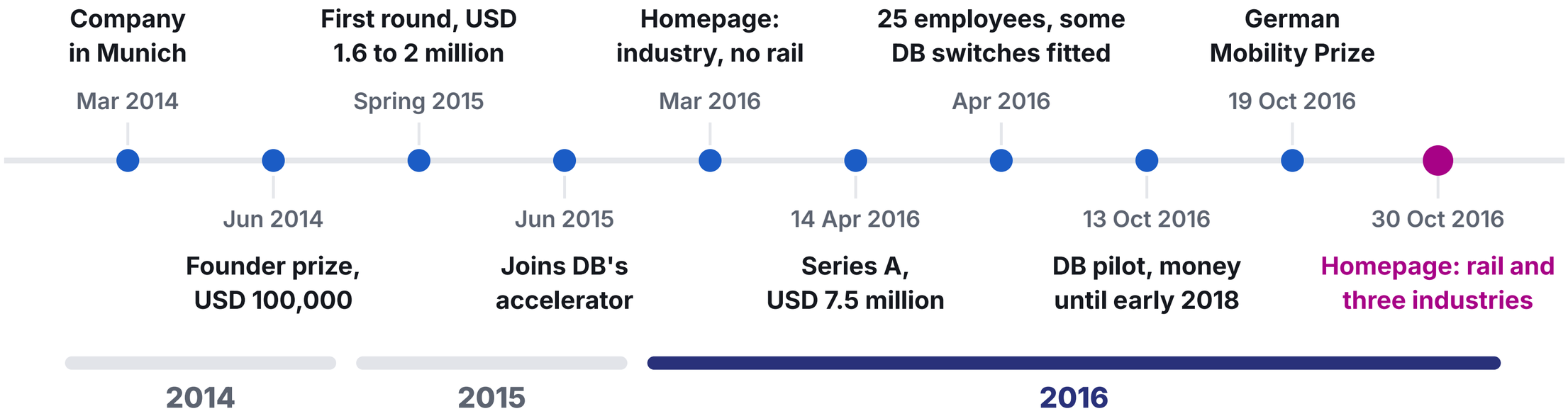
OPTION B

Keep developing Konux for several industries



CASE · WHAT HAPPENED UNTIL 31 OCTOBER 2016

Two and a half years, one decision



DATA Konux case, timeline. Spring 2015: USD 1.6 million (Gründerszene) or 2 million (Hasler, Kunze), as the case reports both.

CASE · EXHIBIT 1

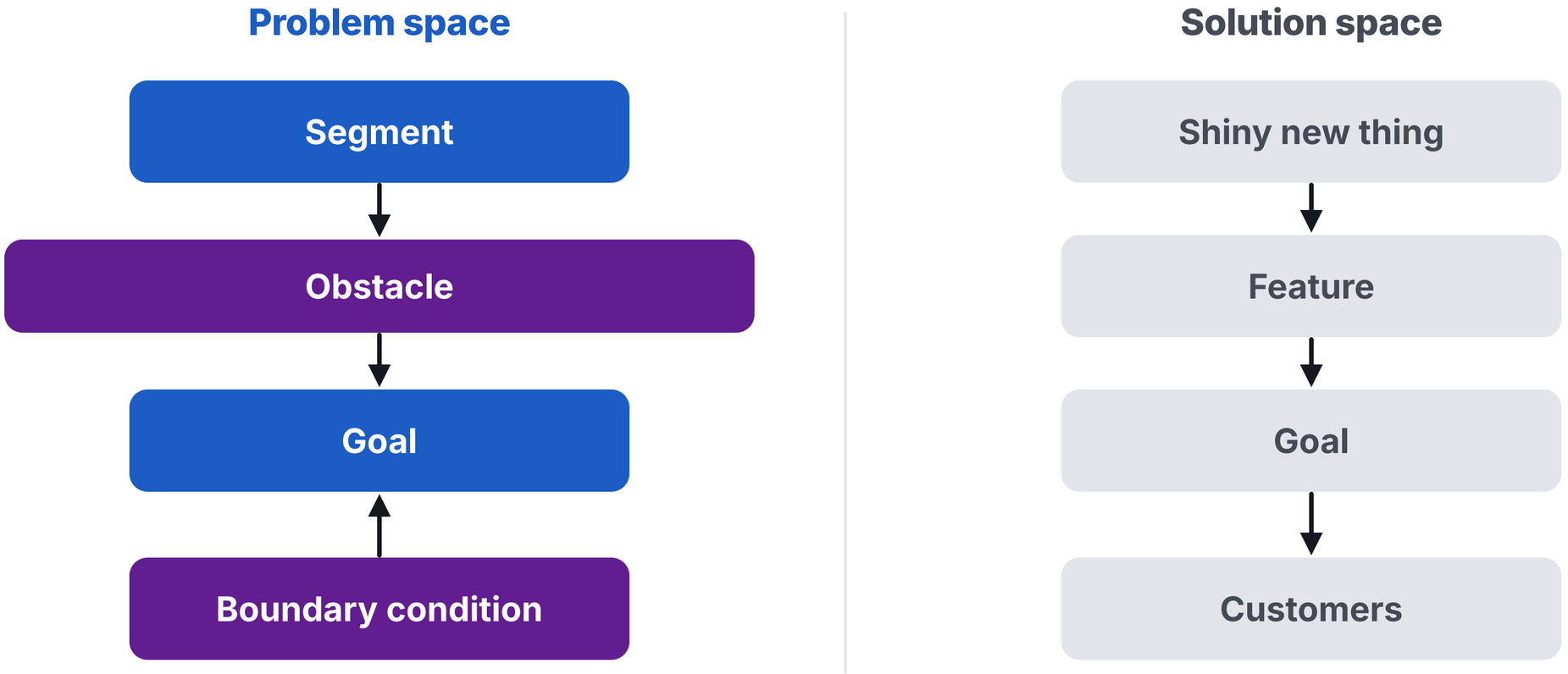
DB Netz switches, 2010 to 2015

	2010	2011	2012	2013	2014	2015
Switches and crossings, DB Netz	71,674	70,477	69,983	69,400	68,516	67,071
Average age of switches, years	19.6	19.6	19.9	20.0	19.9	19.5
Fault reports, all asset groups (priorities 1 to 4)	205,758	195,514	199,953	194,914	183,890	187,338
of which at switches and crossings	36,338	33,025	35,555	35,854	32,152	33,108
Fault duration, all asset groups, minutes (priorities 1 and 2)	8,180,770	7,903,838	8,272,573	9,366,058	9,210,786	9,704,178
of which at switches and crossings	1,377,288	1,309,155	1,449,744	1,630,366	1,604,833	1,725,246
Train delay, all causes, million minutes	159.12	155.48	146.52	164.02	162.43	174.63

DATA Compiled by the case writer from DB Netz AG, Infrastrukturzustands- und -entwicklungsbericht 2014 and 2015 (C37 to C41).

CASE · DIAGNOSIS · 30 MINUTES

Describe DB's problem without a sensor



CASE · SEGMENT TABLE · 30 MINUTES

Which cells did Konux hear?

	How often	How bad	Spend today	Who signs	Heard or assumed
Rail (DB)					
Industrial pumps					
Machine tools					
Elevators					

CASE · VOTE 2 OF 2

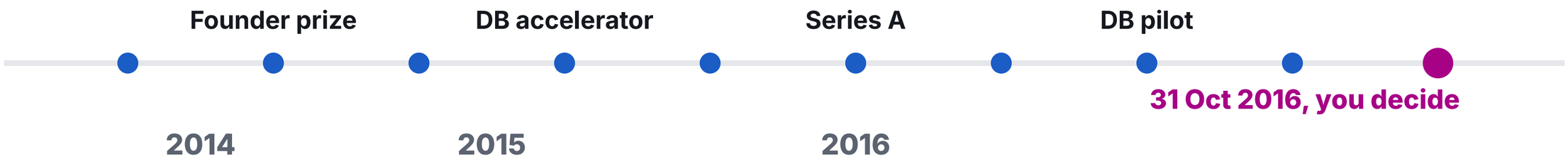
Same question. Who moved, and why?

OPTION A

Build Konux around railway switches

OPTION B

Keep developing Konux for several industries



CASE · BRIDGE

The empty cells are your next questions

Before your next customer conversation, write your own segment table and mark every cell heard or assumed.

	How often	How bad	Spend today	Who signs	Heard or assumed
Your segment 1					
Your segment 2					
Your segment 3					

See you next time!